

KINGMAN, AZ, USA

WMO: 723700

Lat: 35.258N

Lon: 113.933W

Elev: 1042

StdP: 89.41

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 93167

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.4	-2.8	-19.6	0.8	9.5	-17.4	0.9	9.0	13.5	9.1	12.0	9.4	2.3	100	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	39.1	17.8	37.8	17.6	36.8	17.2	20.9	30.9	20.4	30.7	19.8	30.6	6.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	15.3	23.5	17.5	14.2	24.1	16.3	13.2	24.9	65.4	30.6	63.2	30.4	61.0	30.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	11.0	9.6	-8.0	41.8	2.0	1.4	-9.4	42.8	-10.6	43.6	-11.8	44.4	-13.2	45.5
			-9.7	22.8	1.8	2.1	-11.0	24.4	-12.1	25.6	-13.1	26.8	-14.4	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.9	8.4	11.9	15.1	20.3	26.2	29.0	28.3	24.3	17.4	11.1	6.3
	DBStd	8.81	3.38	3.71	3.81	4.30	4.28	3.48	2.35	2.21	3.11	4.02	4.09	3.50
	HDD10.0	366	102	68	26	8	0	0	0	0	0	3	35	123
	HDD18.3	1638	353	279	201	114	31	1	0	0	3	65	217	374
	CDD10.0	2975	8	23	84	160	320	486	590	567	430	232	69	7
	CDD18.3	1206	0	0	1	16	92	237	332	308	183	36	1	0
	CDH23.3	17488	0	3	77	369	1459	3682	4808	4212	2268	567	44	0
	CDH26.7	9701	0	0	11	103	666	2204	2927	2458	1159	171	2	0
Wind	WSAvg	4.2	3.5	3.9	4.3	4.9	5.0	5.2	4.7	4.4	4.0	3.8	3.5	3.3
Precipitation	PrecAvg	251	30	27	32	14	8	6	24	34	20	18	20	25
	PrecMax	474	149	117	122	103	55	38	101	137	75	57	78	91
	PrecMin	100	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	84	31	27	30	22	12	10	25	33	20	17	20	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	23.7	28.0	32.0	36.7	41.0	41.5	39.4	37.1	32.5	26.9	20.2
		MCWB	8.6	9.5	11.4	13.2	15.6	17.7	18.4	18.3	17.2	16.0	11.7	8.9
	2%	DB	18.1	21.3	25.5	29.4	34.2	38.5	39.4	38.2	35.4	30.3	24.6	17.7
		MCWB	7.5	8.6	10.5	12.3	14.5	16.4	18.2	18.3	16.7	14.2	10.6	7.8
	5%	DB	16.3	18.9	23.3	27.4	32.5	37.2	38.0	37.1	34.1	28.6	22.5	15.7
		MCWB	7.0	7.6	9.8	11.5	13.9	15.9	18.0	18.3	16.4	13.2	10.1	6.9
	10%	DB	14.3	16.7	21.5	25.3	30.7	35.9	36.8	35.9	32.7	26.9	20.0	13.4
		MCWB	6.3	7.0	9.1	10.5	13.2	15.3	17.6	18.0	16.3	12.6	9.0	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.9	12.7	14.1	16.6	19.0	21.7	22.3	21.2	18.1	13.7	10.7
		MCDB	16.6	18.5	25.0	30.0	34.3	35.6	30.6	32.8	29.5	26.1	21.7	16.7
	2%	WB	9.3	9.8	11.3	12.9	15.3	17.9	20.9	21.0	20.0	15.9	11.9	9.5
		MCDB	14.4	18.0	23.0	27.7	32.0	34.7	30.5	31.6	28.9	26.3	21.3	15.2
	5%	WB	8.2	8.8	10.4	11.9	14.4	17.0	20.4	20.4	19.1	14.7	10.7	8.2
		MCDB	13.9	17.0	21.7	25.7	31.0	34.2	30.5	31.3	28.0	24.9	20.4	13.8
	10%	WB	7.0	7.7	9.5	10.9	13.5	16.2	19.9	19.8	18.3	13.5	9.7	6.8
		MCDB	13.1	15.1	20.2	24.0	29.5	33.7	30.5	30.9	27.4	24.0	19.4	12.3
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.5	15.1	15.7	15.9	16.4	14.2	14.0	15.0	15.4	14.7	12.7
		MCDBR	16.3	17.4	18.3	18.5	18.1	17.8	16.3	15.6	16.6	17.5	17.7	15.7
		MCWBR	8.9	8.8	8.6	8.3	7.9	7.4	6.2	5.8	6.3	7.8	9.0	8.6
	5% WB	MCDBR	13.0	14.7	16.5	17.2	16.8	16.0	12.5	12.8	12.8	14.3	15.5	12.6
		MCWBR	7.5	7.7	8.2	7.8	7.5	6.8	4.3	4.7	5.3	6.7	8.2	7.4
Clear-Sky Solar Irradiance	taub	0.254	0.259	0.270	0.298	0.305	0.308	0.353	0.360	0.310	0.288	0.259	0.254	
	taud	2.639	2.647	2.593	2.480	2.501	2.490	2.460	2.426	2.550	2.583	2.642	2.644	
	Ebn at Noon	975	1004	1011	987	976	968	923	912	952	953	961	955	
	Edh at Noon	71	79	91	108	107	108	111	113	94	83	70	67	
All-Sky Solar Radiation	RadAvg	3.17	4.12	5.62	7.04	7.95	8.52	7.32	6.78	6.10	4.71	3.56	2.82	
	RadStd	0.26	0.36	0.31	0.30	0.38	0.34	0.41	0.35	0.26	0.29	0.15	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	-2.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	-1.52	N/A	N/A	N/A	N/A	N/A	+104	+46

Nomenclature: See separate page